6GGW / NetSwitch — system report

Generated by ggw_report on this machine. Open on any device, or print to PDF (Ctrl/Cmd+P).

Compute — MFLOPS (double precision)

Dense matrix multiply (512×512, 2·N³ FLOPs)	**3351 MFLOPS**
Sustained FMA loop (2 FLOPs / fma)	**7237 MFLOPS**
Matmul checksum (reproducibility)	`134315886.830217`
FMA checksum (reproducibility)	`4.203541`

MFLOPS = floating-point operations ÷ wall time. Matmul counts 2·N³ FLOPs per pass; FMA counts 2 per fused multiply-add. Same binary run twice on the same box reproduces the checksums.

Processor & memory

CPU	AMD EPYC-Genoa Processor
Logical cores	16
RAM	30.6 GB total / 17.9 GB available
Disk (report volume)	600.7 GB total / 111.6 GB free

Operating system & compatibility

OS	Ubuntu
Version	24.04.3 LTS (Noble Numbat)
Compatibility	Supported (RHEL/CentOS/Rocky/Alma/Fedora/Gentoo/Debian/Ubuntu family)

Supported matrix: RHEL / CentOS / Rocky / AlmaLinux / Fedora / Gentoo / Debian / Ubuntu on Linux; Windows Server 2016 / 2019 / 2022 / 2025 Datacenter on Windows. All modules are single-file C++17 and build from the same sources on both.

How this figure was produced

Method	LINPACK-style dense DGEMM + sustained FMA, wall-clock timed, checksummed so the compiler cannot elide the work.
Repeatability	Deterministic inputs; ≥0.4 s of measured work per figure.
To regenerate	`./ggw_report --n 512`

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